A Novel Bipartite Consensus Tracking Control for Multiagent Systems Under Sensor Deception Attacks

Xinjun Wang, Ye Cao, Ben Niu, and Yongduan Song, *Fellow, IEEE*

***Abstract*—This article presents a novel adaptive bipartite consensus tracking strategy for multiagent systems (MASs) under sensor deception attacks. The fundamental design philosophy is to develop a hierarchical algorithm based on shortest route technology that recasts the bipartite consensus tracking problem for MASs into the tracking problem for a single agent and eliminates the need for any global information of the Laplacian matrix. As the sensors suffer from malicious deception attacks, the states cannot be measured accurately, we thus construct a novel dynamic estimator to estimate the actual states, which, together with a new coordinate transformation involving the attacked and estimated state variables, allows a distributed security control scheme to be developed, in which the singularity of the adaptive iterative process involved in existing works is completely avoided. Furthermore, the Nussbaum functions are included in the controller to account for the influence of the unknown control gains caused by sensor deception attacks. It is shown that the distributed consensus tracking errors converge to a small neighborhood of the origin, and all the signals in the closed-loop system remain bounded. Simulation on a forced damped pendulums (FDPs) is conducted to demonstrate and verify the effectiveness of the proposed strategy.**

***Index Terms*—Adaptive control, bipartite consensus, deception attacks, dynamic estimator, multiagent systems (MASs).**

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I. Introduction

THE PAST decades have witnessed an increasing and wide application of multiple agent systems (MASs) in areas ranging from natural networks to unmanned aerial vehicles, where agents either work cooperatively or act antagonistically [1], [2], [3], [4], [5]. Until now, many meaningful results have been presented in [6], [7], [8], and [9]. A novel distributed optimal coordination was designed for a class of linear MASs [6]. By designing a dynamic threshold-based event-triggered strategy, the consensus tracking control scheme of a class of nonlinear MASs was proposed in [7]. In fact, confrontations among individual agents do exist in real world. For example, in match-play robotic systems, the two team robots might have fierce confrontations and collisions during the game [10]. Thus, the consensus of MASs is no longer limited to the cooperative consensus, and the bipartite consensus issues also play a vital role [11], [12], [13], [14]. The "bipartite" means that all nodes of MASs are divided into two parts, part of the agents attempt to achieve a common goal y_d and the other part of the agents compete to reach a common opposite goal $-y_d$. Some representative results on bipartite consensus control can be found from [11], [12], [13], [14], and [15] and the reference therein. In [15], the problem of bipartite synchronization for a class of coupled competitive neural networks was studied. In [16], a distributed control algorithm was constructed by introducing the Laplacian feedback strategy to solve the problem of bipartite consensus. In [11], [12], and [17], the bipartite consensus control strategy was studied for a class of general linear MASs. But the majority of practical systems are nonlinear and modeled by uncertain dynamics [18], [19]. Thus, developing the distributed bipartite consensus control scheme for nonlinear MASs is more appealing [20], [21], [22], [23], [24], [25], [26], [27]. It should be mentioned that the proposed method in [20], [21], [22], [23], [24], [25], [26], and [27] is based on a common assumption that the condition of structurally balanced for a signed digraph should be satisfied, and the global information of the Laplacian matrix is required, thus when a signed digraph is unbalanced, how to get the bipartite consensus without using the Laplacian-like strategies requires further investigation.

With the rapid development of information fusion and networked system, cyber security has attracted more and more attention [28], [29], [30], [31], [32], [33], [34], [35], [36]. In [30], the secure state estimation strategy was proposed

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and successfully applied to the multiagent network system. In normal conditions, the cyber attacks models can be divided into two types, that is, denial-of-service (DOS) attacks [37] and deception attacks [38]. The DOS attacks try to prevent the legitimate user from using their required resources, such as bandwidth and computational resources. The purpose of deception attacks is to influence the data integrity by the injection of false information into sensors. Among them, the deception attacks are more practical and more reachable compared with the DOS attacks in an insecure network environment, which might destroy satisfactory consensus performance. Therefore, it is necessary to develop the security control strategy subject to unknown deception attacks. Until now, a series of meaningful research works about the network security problems have been devoted for different systems [38], [39], [40], [41]. In [39], the consensus control issue was discussed for leaderless MASs suffering from sensor and actuator attacks. The secure synchronization of MASs under deception attacks in the impulsive control framework was found in [40]. It should be mentioned that for the security problems of CPSs suffering from malicious attacks, since the state signals are damaged by attacks, the true states of the system cannot be obtained and employed directly to design a distributed adaptive control strategy. Thus, realizing the bipartite tracking control for the MASs with false data injection attacks is a challenging task since the traditional error transformation methods used in [20], [21], [22], [23], [24], [25], [26], [27], [42], and [43] are not suitable for the controller design and only the compromised states are available. As far as the authors know, there are no results in the bipartite tracking consensus control for nonlinear MASs against deception attacks under a structurally unbalanced digraph.

This study investigates the adaptive bipartite consensus tracking problem for MASs under the sensor deception attacks. First, with the help of the hierarchical algorithm and the bipartition process, the condition of structurally unbalanced for a signed digraph is considered, and the consensus tracking problem is transformed into the tracking problem of each agent. Then, as the sates measured by sensors are attacked by malicious deception attacks, we construct a novel distributed security control scheme by using the attacked and estimated state signals. Finally, by introducing the Nussbaum functions, the technical difficulty in control design and stability analysis arising from unknown and time-varying control gains is circumvented. The main contributions of our work are as follows.

1) Since the true states of the system are not available for feedback design, a novel dynamic estimator employing the compromised state information is designed. On this basis, a new coordinate transformation involving the attacked and estimated state variables is constructed, with which a recursive control design strategy capable of achieving bipartite consensus tracking for the MASs under deception attacks is developed.
2) Different from the existing bipartite consensus results that rely on the Laplacian-like matrix [20], [21], [22], [23], [24], [25], [26], [27], [42], [43], the proposed hierarchical algorithm is based on the shortest route technology, thus it allows the consensus tracking problem for multiagents to be decomposed into the tracking problem for a single agent, and the global Laplacian matrix is not required, rendering the strategy fully distributed.
3) By introducing a new auxiliary variable, the singularity problem in the adaptive iteration process of the existing work [44] is avoided. Moreover, unlike the work [38] requiring that the signs of the attack gains be known a priori, the proposed method, with the aid of the Nussbaum function, does not need to know the signs of the attack gains in its design.

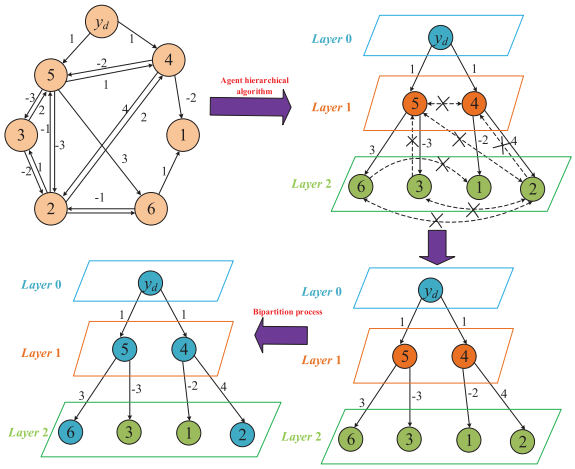


Fig. 1. Hierarchical decomposition of agents.

II. System Descriptions

A. Graph Theory

A weighted signed digraph $G \triangleq \{O, E, A\}$ is introduced to denote the interactions between agents, where $O = \{o_i : 1, \ldots, N\}$ represents a set of follower agents, N is the number of agents, $E \subseteq \{e_{ij} : i = 1, \ldots, N, j = 1, \ldots, N, i \neq j\}$ denotes a set of edges, and adjacency matrix is described as $A = [a_{ij}] \in R^{N\times N}$. When $a_{ij} < 0$, the relationship between o_i and o_j is antagonistic; and $a_{ij} > 0$ if relationship between o_i and o_j be cooperative interaction; otherwise $a_{ij} = 0$. $N_i = \{j \mid a_{ij} \neq 0\}$ represents all the neighbors of node i. The Laplacian matrix is defined as $L = D - A$, where $L \in R^{N\times N}$, $D = \text{diag}(d_i)$ is the in-degree matrix of the ith follower, and $d_i = \sum_{j\in N_i} |a_{ij}|$. In this work, due to considering the leader agent y_d, a signed digraph $\bar{G} \triangleq \{\bar{O}, \bar{E}, \bar{A}\}$ is also introduced to denote the interactions between the leader and the followers, where $\bar{G} = G \cup \{y_d\}$ and $\bar{E} = \bar{G} \cup \bar{G}$.

B. Hierarchical Algorithm

In this article, the signed digraph only has one directed spanning tree, and the leader is regarded as root node. In order to avoid the problem of algebraic loop, the hierarchical algorithm (as shown in Table I) and the bipartition process (as shown in Table II) are used here. Now, we describe the crucial idea of the hierarchical algorithm and bipartition process. In the hierarchical algorithm, only one node's information in the upper layer is transmitted for each agent in the lower

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TABLE I
HIERARCHICAL ALGORITHM

Algorithm 1: Hierarchical algorithm of agents with r layers ($1 \le r \le N$)

(i) Initialization: 'Layer 0' -> Leader y_d;
(ii) Layer 1**: if** the agent o_i in G can receive the information from y_d directly, i.e., $y_d(t)$ -> o_i;
then $o_i \in$ **Layer 1**;
(iii) Layer q**: if** the agent o_i in G can receive the information from Layer $q-1$, i.e., 'Layer $q-1$' -> o_i;
then $o_i \in$ **Layer q** ($q = 2, \ldots, r$);
(iv) For the agent $o_i \in$ Layer q, find a unique agent $o_j \in$ Layer $q-1$ in the upper layer that agent o_i only require obtaining o_j's information: o_j is chosen as $\max\{|a_{i,j}|\} \to o_j$.
(v) Each agent only require obtaining its unique node's information in the upper layer.

TABLE II
BIPARTITION PROCESS BY APPLYING HIERARCHICAL ALGORITHM

Algorithm 2: Bipartition process of agents

step1: $\forall o_i$ in G, find its only identified node o_p;
step2: Determine the shortest path S_i:
$y_d \to o_m \to \cdots o_p \to o_i$;
step3: Calculate $a_{i,p} \cdots a_{m,d}$;
step4: Define two sets $M_1 = \{o_i | S_i > 0\}$ and $M_2 = \{o_i | S_i < 0\}$;
step5: If $a_{i,p} \cdots a_{m,d} > 0$, then $S_i > 0$, $o_i \in M_1$; else $S_i < 0$, $o_i \in M_2$;
step6: Repeat the above step1 - step5.

layer. In Fig. 1, agent $o_i, i = 1, \ldots, 6$ consist of three layers "layer 0 - layer 2" according to Algorithm 1. Suppose that o_i can receive the information from its only identified node o_p, and a shortest route S_i is determined as $y_d \to o_m \to \cdots o_p \to o_i$. When edge weights $a_{i,p}, \ldots, a_{m,d} > 0$, one has $S_i > 0$, else $S_i < 0$. Then, we define two sets M_1 and M_2. If $S_i > 0$, $o_i \in M_1$ holds. Otherwise, $o_i \in M_2$ holds. By repeating the above process, a bipartition is successfully achieved in Algorithm 2. For example, as shown in Fig. 1, $o_2, o_4, o_5, o_6 \in M_1$, and $o_1, o_3 \in M_2$.

C. Model Description

Consider the ith follower agent with the following dynamics:

$$\begin{aligned} \dot{x}_{i,k} &= x_{i,k+1} + f_{i,k}(\bar{x}_{i,k}) \\ \dot{x}_{i,n} &= u_i + f_{i,n}(x_i) \\ \check{x}_{i,k} &= x_{i,k} + \omega_s(t, x_{i,k}) \\ y_i &= x_{i,1} \end{aligned} \tag{1}$$

where $i = 1, \ldots, N$, $k = 1, \ldots, n-1$; $\bar{x}_{i,k} = [x_{i,1}, x_{i,2}, \ldots, x_{i,k}]^T \in R^k$ and $x_i = [x_{i,1}, x_{i,2}, \ldots, x_{i,n}]^T \in R^n$ stand for the system states of the ith follower, $y_i \in R$ and $u_i \in R$ are the system output and protocol, respectively. $f_{i,k}(\cdot)$ is the unknown smooth function. $\omega_s(t, x_{i,k})$ is the sensor deception attack and $\check{x}_{i,k}$ is the available system state after attacks.

The main objectives of this article are to design a distributed security control scheme u_i for each follower agent (1) suffering from unknown sensor deception attacks such that:

1) the bipartite tracking is achieved, that is, $y_i \to y_d$, $i \in M_1$ or $y_i \to -y_d$, $i \in M_2$;
2) all the signals of the closed-loop systems are bounded.

To achieve the above-mentioned control objectives, the following assumptions and definition are needed.

Assumption 1: The signed digraph G only has a directed spanning tree, and the leader is regarded as root node.

Assumption 2: The error $\omega_s(t, x_{i,k})$ caused by sensor deception attacks are defined as $\omega_s(t, x_{i,k}) = \rho(t)x_{i,k}(t)$ with unknown time-varying signal $\rho(t)$. Besides, $1 + \rho(t) \neq 0$ holds, and there exist unknown constants $\bar{\rho}$ and $\bar{\rho}_d$ such that $|\rho(t)| \le \bar{\rho}$ and $|\dot{\rho}(t)| \le \bar{\rho}_d$.

Definition 1 [44]: For the continue function $N(b) : R \to R$, if the following equalities hold:

$$\lim_{b\to\infty} \sup \frac{1}{b} \int_0^b N(s)ds = +\infty \tag{2}$$

$$\lim_{b\to\infty} \inf \frac{1}{b} \int_0^b N(s)ds = -\infty \tag{3}$$

then function $N(\cdot)$ is a Nussbaum-type function.

There exist lots of functions classified as the Nussbaum-type functions, such as $e^2\cos((\pi/2)\theta)$ and $\theta^2\cos(\theta)$. In this article, the Nussbaum-type function is defined as $N(\theta_l) = e^{\theta_l^2}\sin(\pi\theta_l)$.

Lemma 1 [44]: Define two smooth functions $V(t) : R_{\ge 0} \to R_{\ge 0}$ and $\theta_m(t) : R_{\ge 0} \to R_{\ge 0}$. $N(\cdot)$ is a Nussbaum-type function. Let the time-varying function $g_m(t) \in G = [g_m^-, g_m^+]$ with $0 \notin G$ and the bounds satisfy $g_m^- g_m^+ > 0$. If there exist two positive constants C and D so that

$$\dot{V}(t) \le -CV + D + \sum_{m=1}^{n} \big(g_m(t)N(\theta_m(t)) + 1\big)\dot{\theta}_m(t) \tag{4}$$

$$\dot{\theta}_m(t) \ge 0 \quad \forall t > 0 \tag{5}$$

hold, then $V(t)$ and $\theta_m(t)$ are bounded over $[0, \infty)$.

Remark 1: In comparison with the existing bipartite consensus tracking results in [11] and [45], where the structurally balanced for the digraph is needed, here the condition for digraph is relaxed in Assumption 1. Since the hierarchical algorithm is utilized in this article and each agent only requires obtaining its unique node's information in the upper layer, the computing burden and the communication burden are greatly alleviated in our proposed method. In addition, Assumption 2 is common for the systems with deception attacks, which can be found in [46] and [47] and the references therein. From the definition of coordinate transformation, we can get that $\check{x}_{i,k} = (1 + \rho(t))x_{i,k}$. Thus, the condition $1 + \rho(t) \neq 0$ is to avoid that no information of state signals $x_{i,k}$ is available for control design, causing uncontrollable of the system.

D. Neural Networks

In this article, the radial basis function neural networks (RBFNNs) are applied to approximate the unknown terms in

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the system. The RBFNNs are characterized by as follows:

$$f(Z) = \Phi^{*T}\xi(Z) + \delta(Z) \tag{6}$$

where $\Phi^* = \arg\min_{\Phi\in R^l}\{\sup_{Z\in\Omega_Z}|f(Z) - \Phi^T\xi(Z)|\}$ and $Z \in R^q$ is input vector, $\Phi = [\phi_1, \phi_2, \ldots, \phi_l]^T$ represents weight vector with $l > 1$ being the node number. $\xi(Z) = [\xi_1(Z), \ldots, \xi_l(Z)]^T$ denotes the basis function vector, $\delta(Z)$ denotes approximation error. The basis functions are usually selected as the Gaussian function in the following form:

$$\xi_j(Z) = \exp\Big(-\big(Z - c_j\big)^T\big(Z - c_j\big)/2b_j^2\Big), j = 1, 2, \ldots, l \tag{7}$$

where b_j and c_j denote the widths and centers of the Gaussian function.

Lemma 2 [48]: Let $\bar{x}_q = [x_1, \ldots, x_k, \ldots, x_q]^T$ and $\bar{x}_k = [x_1, \ldots, x_k]^T$. $\xi(\bar{x}_k) = [\xi_1(\bar{x}_k), \ldots, \xi_l(\bar{x}_k)] \in R^l$ and $\xi(\bar{x}_q) = [\xi_1(\bar{x}_q), \ldots, \xi_l(\bar{x}_q)] \in R^l$ be the basis function vectors of RBFNNs. Then, for any positive integer $k \le q$, the inequality $\|\xi(\bar{x}_q)\|^2 \le \|\xi(\bar{x}_k)\|^2$ holds.

Lemma 3 [49]: For the basis function vectors (7) in RBFNNs, $\|\xi(Z)\|^2 \le l$ holds, where l is the node number.

Assumption 3: Assume that Φ and $\delta(Z)$ are unknown but bounded by $\|\Phi\| \le \bar{\Phi}$ and $|\delta(Z)| \le \bar{\delta}$ with $\bar{\Phi}$ and $\bar{\delta}$ being unknown positive constants.

III. Main Results

In this article, since the sensor suffers from the deception attacks and the actual states of the system are unavailable, a novel state estimator involving the compromised information of the states is constructed. Based on the hierarchical algorithm, the bipartite tracking error is decomposed into the sum of the agents' output errors that transmit information within the adjacent layer. Then, a distributed security control scheme is presented by using the new coordinate transform method and the Nussbaum-type functions. The main results are shown as follows.

A. Design of Novel State Estimator

For the CPSs with sensor deception attacks, the states information of the system is damaged. Thus, the "original states" (actual states) $x_{i,k}$ are unavailable and only "compromised states" (states with deception attacks) $\check{x}_{i,k}$ can be obtained. Thus, a novel state estimator is constructed by using the information of compromised states as follows:

$$\begin{aligned}\dot{\hat{x}}_i &= A_{i0}\hat{x}_i + D_i u_i - L_i\check{x}_{i,1}\\ \hat{y}_i &= C_i^T\hat{x}_i\end{aligned} \tag{8}$$

where $\hat{\bar{x}}_i = [\hat{x}_{i,1}, \ldots, \hat{x}_{i,n}]^T$ is the estimate of x_i, $D_i = [0, 0, \ldots, 1]^T$, $C_i^T = [1, 0, \ldots, 0]$, $A_{i0} = A_i + L_iC_i^T$, $L_i = [-l_{i,1}, -l_{i,2}, \ldots, -l_{i,n}]^T$, and

$$A_i = \begin{bmatrix} 0 & 1 & & & \\ 0 & 0 & 1 & & \\ \vdots & \vdots & & \ddots & \\ 0 & 0 & \ldots & 0 & 1 \\ 0 & 0 & \cdots & 0 & 0 \end{bmatrix} \tag{9}$$

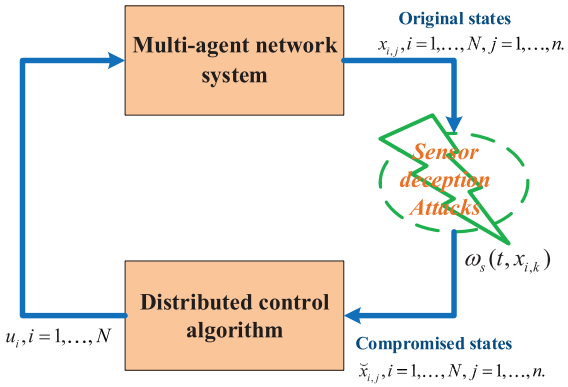


Fig. 2. Controlled nonlinear MASs with sensor deception attacks.

L_i is the design matrix such that the matrix A_{i0} is a strict Hurwitz matrix. At the same time, there exists a positive definite matrix $P_i > 0$ such that

$$A_{i0}^TP_i + P_iA_{i0} = -Q_i \tag{10}$$

with $Q_i > 0$ being a given matrix. Define the estimation errors as $e_{i,q} = x_{i,q} - \hat{x}_{i,q}$. Then, we can obtain the observer error equation as

$$\dot{e}_i = A_{i0}e_i + F(x_i) + L_i\omega_s\big(t, x_{i,1}\big) \tag{11}$$

where $e_i = [e_{i,1}, \ldots, e_{i,n}]^T$, $F(x_i) = [f_{i,1}(\bar{x}_{i,1}), \ldots, f_{i,n}(\bar{x}_i)]^T$.

B. Distributed Adaptive Control Design

This section will be devoted to solving the problem of distributed consensus control for the MASs (1) suffering from sensor deception attacks. For a given node o_i ($o_i \in$ Layer $k, k \ge 1$), the bipartite tracking error between o_i and the leader y_d is defined as $\Lambda_i = x_{i,1} - \text{sign}(S_i)y_d$. Since the hierarchical algorithm is utilized here and o_i can only receive the information from its identified node o_q in the upper layer, thus the consensus tracking error Λ_i is decomposed into the sum of the nodes' output errors, where these nodes should within the adjacent layers and can transmit information from the upper layer to the lower layer. Thus, the error between o_i and o_q is first analyzed and the new error transformations are defined as

$$\begin{aligned} z_{i,1} &= x_{i,1} - \text{sign}\big(a_{i,q}\big)x_{q,1}\\ z_{i,k} &= \lambda\hat{x}_{i,k} - \lambda\alpha_{i,k-1}, k = 2, \ldots, n\end{aligned} \tag{12}$$

where $a_{i,q}$ is the edge weight between agent o_i and o_q, $\alpha_{i,k-1}$, $(k = 2, 3, \ldots, n)$ are virtual control signals to be constructed and $\lambda = (1 + \rho)^{-1}$. According to Assumption 2, we have $0 < \lambda_m \le |\lambda| \le \lambda_M$ and $|\dot{\lambda}| \le \bar{\dot{\lambda}}$, where λ_m, λ_M, and $\bar{\dot{\lambda}}$ are unknown positive constants.

As shown in Fig. 2, the original state signals $x_{i,k}(k = 1, \ldots, n)$ of system are maliciously damaged by the attackers, the error surfaces $z_{i,1}$ and $z_{i,k}$ in (12) cannot be applied to the design of virtual and actual control. Hence, the new compromised coordinate transformation mechanisms need be constructed for control design. Here, new coordinate

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transformation of the error is proposed in the framework of recursive design

$$\begin{aligned}\check{z}_{i,1} &= \check{x}_{i,1} - \operatorname{sign}(a_{i,q})\check{x}_{q,1}\\ \check{z}_{i,k} &= \hat{x}_{i,k} - \alpha_{i,k-1}, \quad k = 2, \ldots, n\end{aligned} \tag{13}$$

where $\check{x}_{i,k}(k = 1, \ldots, n)$ and $\check{x}_{q,1}$ are available state information after attacks, which have been defined in (1).

By recalling Assumption 2 and the definition of λ, the formula $x_{i,k} = \lambda\check{x}_{i,k}$ holds. Thus, by combining the definition of $z_{i,k}$ and $\check{z}_{i,k}$ in (12) and (13), it is easy to get that $z_{i,k} = \lambda\check{z}_{i,k}$, and the compromised error variables $\check{z}_{i,k}$ will be used to construct the controller. The following control design follows the backstepping design procedure through which the feedback control is recursively derived at the final step n.

Step 1: To start with the control design, we introduce a constant

$$\theta_i = \max\{\phi_{i,j} : j = 0, 1, 2, \ldots, n\} \tag{14}$$

where θ_i is an unknown constant, $\hat{\theta}_i$ is the estimate of θ_i, and $\tilde{\theta}_i = \theta_i - \hat{\theta}_i$ is the estimation error of θ_i, $\phi_{i,j}$ is an unknown constant related with the norm of the NN weight vector which will be defined later. The Lyapunov function $V_{i,1}$ is chosen as

$$V_{i,1} = e_i^T P_i e_i + \frac{\lambda^2}{2\lambda_M} z_{i,1}^2 + \frac{1}{2}\tilde{\theta}_i^2 + \frac{1}{2}\tilde{\zeta}_i^2 \tag{15}$$

where $\tilde{\zeta}_i = \zeta_i - \hat{\zeta}_i$ is the estimation error, $\hat{\zeta}_i$ is the estimation of ζ_i, and $\zeta_i = \lambda_M^2\bar{\lambda} + \lambda_M^4$.

The derivative of $V_{i,1}$ is given as

$$\begin{aligned}\dot{V}_{i,1} &= -e_i^T Q_i e_i + 2e_i^T P_i F + 2e_i^T P_i L_i \omega_s(t, x_{i,1}) + \frac{\lambda^2}{\lambda_M} z_{i,1}\\ &\quad \times \left(\check{z}_{i,2} + \alpha_{i,1} + e_{i,2} + f_{i,1} - \operatorname{sign}(a_{i,j})(x_{q,2} + f_{q,1})\right)\\ &\quad + \frac{\lambda\dot{\lambda}}{\lambda_M} z_{i,1}^2 + \tilde{\theta}_i\dot{\tilde{\theta}}_i + \tilde{\zeta}_i\dot{\tilde{\zeta}}_i\\ &\le -e_i^T Q_i e_i + 2e_i^T P_i F + 2e_i^T P_i L_i \omega_s(t, x_{i,1}) + \zeta_i\check{z}_{i,1}^2\\ &\quad + \frac{z_{i,1}^2}{2} + \frac{z_{i,2}^2}{2} + \|e_i\|^2 + \frac{\lambda^2}{\lambda_M} z_{i,1}(\alpha_{i,1} + \bar{f}_{i,1})\\ &\quad + \tilde{\theta}_i\dot{\tilde{\theta}}_i + \tilde{\zeta}_i\dot{\tilde{\zeta}}_i\end{aligned} \tag{16}$$

where $\bar{f}_{i,1} = f_{i,1} - \operatorname{sign}(a_{i,q})(x_{q,2} + f_{q,1})$. Since the nonlinear function $f_{i,k}(\bar{x}_{i,k})$, $k = 1, 2, \ldots, n$ contains unknown functions, thus the RBFNN is introduced to approximate the function, that is, $f_{i,k}(\bar{x}_{i,k}) = \Phi_{i,k0}^T\xi_{i,0}(X_{i,0}) + \delta_{i,k0}(X_{i,0})$, $\delta_{i,k0}$ is the approximation error, $|\delta_{i,k0}| \le \bar{\delta}_{i,k0}$. Then, we have

$$F(x_i) = \Phi_{i,0}^T\xi_{i,0}(X_{i,0}) + \delta_{i,0}(X_{i,0}) \tag{17}$$

where $X_{i,0} = x_i \in \Omega_{i,0}$, and $\Omega_{i,0}$ is a compact set for x_i, $\Phi_{i,0} = (\Phi_{i,10}^T, \ldots, \Phi_{i,n0}^T)$, $\delta_{i,0}(X_{i,0}) = (\delta_{i,10}(X_{i,0}), \ldots, \delta_{i,n0}(X_{i,0}))^T$, $\|\delta_{i,0}(X_{i,0}\| \le \bar{\delta}_{i,0}$. By using Young's inequality, the following inequalities hold:

$$\begin{aligned}2e_i^T P_i F &= 2e_i^T P_i(\Phi_{i,0}^T\xi_{i,0}(X_{i,0}) + \delta_{i,0}(X_{i,0}))\\ &\le 2\|e_i\|^2 + \|P_i\|^2\phi_{i,0} + \|P_i\|^2\bar{\delta}_{i,0}^2\\ &\le 2\|e_i\|^2 + \|P_i\|^2\theta_i + \|P_i\|^2\bar{\delta}_{i,0}^2\end{aligned} \tag{18}$$

$$2e_i^T P_i L_i \omega_s(t, x_{i,1}) \le \|e_i\|^2 + \|P_i\|^2\|L_i\|^2\bar{\omega}_1^2 \tag{19}$$

where $\phi_{i,0} = \|\Phi_{i,0}\|^2 l_{i,0}$, and by using Lemma 3, $\xi_0^T(X_0)\xi_0(X_0) \le l_{i,0}$ with $l_{i,0}$ being a node number, and $\omega_s(t, x_{i,1}) \le \bar{\omega}_1$. Similarly, $\bar{f}_{i,1} = \Phi_{i,1}^T\xi_{i,1}(M_{i,1}) + \delta_{i,1}$, by using Lemma 2, we have $\bar{f}_{i,1} \le \Phi_{i,1}^T\xi_{i,1}(\bar{M}_{i,1}) + \delta_{i,1}$, then, based on Young's inequality, the following inequality holds:

$$\begin{aligned}\frac{\lambda^2}{\lambda_M} z_{i,1}\bar{f}_{i,1} &= \frac{\lambda^2}{\lambda_M} z_{i,1}(\Phi_{i,1}^T\xi_{i,1}(M_{i,1}) + \delta_{i,1})\\ &\le \frac{1}{2}\check{z}_{i,1}^2\phi_{i,1}\xi_{i,1}^T(\bar{M}_{i,1})\xi_{i,1}(\bar{M}_{i,1}) + \frac{1}{2} + \frac{z_{i,1}^2}{2} + \frac{\lambda_M^2\bar{\delta}_{i,1}^2}{2}\\ &\le \frac{1}{2}\check{z}_{i,1}^2\theta_i\xi_{i,1}^T(\bar{M}_{i,1})\xi_{i,1}(\bar{M}_{i,1}) + \frac{1}{2} + \frac{z_{i,1}^2}{2} + \frac{\lambda_M^2\bar{\delta}_{i,1}^2}{2}\end{aligned} \tag{20}$$

where $M_{i,1} = [x_{i,1}, x_{q,1}, x_{q,2}]^T$, $\bar{M}_{i,1} = [x_{i,1}, x_{q,1}]^T$, $\phi_{i,1} = \lambda_M^4\|\Phi_{i,1}\|^2$, $\delta_{i,1}$ is the approximation error, $|\delta_{i,1}| \le \bar{\delta}_{i,1}$.

Then, we have

$$\begin{aligned}\dot{V}_{i,1} &\le -(\lambda_{\min}(Q_i) - 4)\|e_i\|^2 + \|P_i\|^2\theta_i + \|P_i\|^2\bar{\delta}_{i,0}^2\\ &\quad + \|P_i\|^2\|L_i\|^2\bar{\omega}_1^2 + \frac{\lambda^2}{\lambda_M} z_{i,1}\alpha_{i,1} + z_{i,1}^2 + \frac{z_{i,2}^2}{2}\\ &\quad + \frac{1}{2}\check{z}_{i,1}^2\theta_i\xi_{i,1}^T(\check{M}_{i,1})\xi_{i,1}(\check{M}_{i,1}) + \check{z}_{i,1}^2\chi_{i,1}\\ &\quad + \frac{1}{2} + \frac{\lambda_M^2\bar{\delta}_{i,1}^2}{2} + \tilde{\theta}_i\dot{\tilde{\theta}}_i + \tilde{\zeta}_i\dot{\tilde{\zeta}}_i\end{aligned} \tag{21}$$

where $\chi_{i,1} = (1/2)\theta_i(\xi_{i,1}^T(\bar{M}_{i,1})\xi_{i,1}(\bar{M}_{i,1}) - \xi_{i,1}^T(\check{M}_{i,1})\xi_{i,1}(\check{M}_{i,1}))$ with $\check{M}_{i,1} = [\check{x}_{i,1}, \check{x}_{q,1}]^T$.

The virtual controller $\alpha_{i,1}$ is chosen as

$$\alpha_{i,1} = -k_{i,1}\check{z}_{i,1} - \check{z}_{i,1} + N(\tau_{i,1})\Psi_{i,1} \tag{22}$$

with

$$\begin{aligned}\Psi_{i,1} &= \frac{1}{2}\check{z}_{i,1}\hat{\theta}_i\xi_{i,1}^T(\check{M}_{i,1})\xi_{i,1}(\check{M}_{i,1}) + \check{z}_{i,1}\hat{\zeta}_i + \frac{1}{2}\check{z}_{i,1}^3,\\ \dot{\tau}_{i,1} &= \check{z}_{i,1}\Psi_{i,1}, \dot{\hat{\zeta}}_i = \check{z}_{i,1}^2 - r\hat{\zeta}_i.\end{aligned} \tag{23}$$

Substituting (22) and (23) into (21) results in

$$\begin{aligned}\dot{V}_{i,1} &\le -(\lambda_{\min}(Q_i) - 4)\|e_i\|^2 + \|P_i\|^2\theta_i + \|P_i\|^2\bar{\delta}_{i,0}^2\\ &\quad + \|P_i\|^2\|L_i\|^2\bar{\omega}_1^2 - k_{i,1}z_{i,1}^2 + \frac{1}{2}z_{i,2}^2 + (g_{i,1}(t)N + 1)\dot{\tau}_{i,1}\\ &\quad + \frac{1}{2}\chi_{i,1}^2 + \frac{1}{2} + \tilde{\theta}_i(\pi_{i,1} - \dot{\hat{\theta}}_i) + k_{i,0}\hat{\theta}_i\tilde{\theta}_i + r\hat{\zeta}_i\tilde{\zeta}_i + \frac{\lambda_M^2\bar{\delta}_{i,1}^2}{2}\end{aligned} \tag{24}$$

where $g_{i,1}(t) = \lambda^3/\lambda_M$, and

$$\pi_{i,1} = \sigma_{i,1} - k_{i,0}\hat{\theta}_i, \sigma_{i,1} = \frac{1}{2}\check{z}_{i,1}^2\xi_{i,1}^T(\check{M}_{i,1})\xi_{i,1}(\check{M}_{i,1}). \tag{25}$$

Step 2: Consider the following Lyapunov function candidate:

$$V_{i,2} = V_{i,1} + \frac{1}{2}z_{i,2}^2 + \frac{1}{2}\tilde{\gamma}_i^2 \tag{26}$$

where $\tilde{\gamma}_i = \gamma_i - \hat{\gamma}_i$ is the estimation error, $\hat{\gamma}_i$ is the estimation of γ_i and $\gamma_i = \bar{\lambda}^2\lambda_M^2$. It follows that:

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$$\dot{V}_{i,2} = \dot{V}_{i,1} + \tilde{\gamma}_i\dot{\hat{\gamma}}_i + z_{i,2}\Bigg(z_{i,3} + \lambda\alpha_{i,2} + \lambda l_{i,2}\big(\check{x}_{i,1} - \hat{x}_{i,1}\big) + \dot{\lambda}\hat{x}_{i,2} - \dot{\lambda}\alpha_{i,1} - \lambda\Bigg(\frac{\partial\alpha_{i,1}}{\partial\tau_{i,1}}\dot{\tau}_{i,1} + \frac{\partial\alpha_{i,1}}{\partial\hat{\theta}_i}\dot{\hat{\theta}}_i + \frac{\partial\alpha_{i,1}}{\partial\hat{\zeta}_i}\dot{\hat{\zeta}}_i + \frac{\partial\alpha_{i,1}}{\partial\check{x}_{i,1}}\frac{\lambda\dot{x}_{i,1} - \dot{\lambda}x_{i,1}}{\lambda^2} + \frac{\partial\alpha_{i,1}}{\partial\check{x}_{q,1}}\frac{\lambda\dot{x}_{q,1} - \dot{\lambda}x_{q,1}}{\lambda^2}\Bigg)\Bigg). \tag{27}$$

Define $G_2 = \alpha_{i,1} - ([\partial\alpha_{i,1}]/[\partial\check{x}_{i,1}])\check{x}_{i,1} - ([\partial\alpha_{i,1}]/[\partial\check{x}_{q,1}])\check{x}_{q,1} + \hat{x}_{i,2}$. Note that $z_{i,2}\dot{\lambda}G_2 = \check{z}_{i,2}\lambda\dot{\lambda}G_2 \le (1/2)\check{z}_{i,2}^2\bar{\lambda}^2\lambda_M^2G_2^2 + 1/2$. On the other hand, similar to (20), we have $z_{i,2}([\partial\alpha_{i,1}/\partial\check{x}_{i,1}]f_{i,1} + [\partial\alpha_{i,1}/\partial\check{x}_{q,1}]f_{q,1}) \le (1/2)\check{z}_{i,2}^2\phi_{i,2}\xi_{i,2}^T(\bar{M}_{i,2})\xi_{i,2}(\bar{M}_{i,2}) + (1/2) + (z_{i,2}^2/2) + (\bar{\delta}_{i,2}^2/2) \le (1/2)\check{z}_{i,2}^2\theta_i\xi_{i,2}^T(\bar{M}_{i,2})\xi_{i,2}(\bar{M}_{i,2}) + (1/2) + (z_{i,2}^2/2) + (\bar{\delta}_{i,2}^2/2)$ with $\bar{M}_{i,2} = [x_{i,1}, x_{i,2}, x_{q,1}]^T$ and $\phi_{i,2} = \lambda_M^2\|\Phi_{i,2}\|^2$, $\delta_{i,2}$ is the approximation error, $|\delta_{i,2}| \le \bar{\delta}_{i,2}$.

The virtual controller $\alpha_{i,2}$ is chosen as

$$\alpha_{i,2} = -k_{i,2}\check{z}_{i,2} - \frac{3}{2}\check{z}_{i,2} + \frac{\partial\alpha_{i,1}}{\partial\tau_{i,1}}\dot{\tau}_{i,1} + \frac{\partial\alpha_{i,1}}{\partial\hat{\zeta}_i}\dot{\hat{\zeta}}_i + \frac{\partial\alpha_{i,1}}{\partial\hat{\theta}_i}\pi_{i,2} + \frac{\partial\alpha_{i,1}}{\partial\check{x}_{i,1}}\check{x}_{i,2} + \frac{\partial\alpha_{i,1}}{\partial\check{x}_{q,1}}\check{x}_{q,2} - l_{i,2}\big(\check{x}_{i,1} - \hat{x}_{i,1}\big) + N\big(\tau_{i,2}\big)\Psi_{i,2} \tag{28}$$

with

$$\Psi_{i,2} = \frac{1}{2}\check{z}_{i,2}\hat{\theta}_i\xi_{i,2}^T\Big(\check{M}_{i,2}\Big)\xi_{i,2}\Big(\check{M}_{i,2}\Big) + \frac{1}{2}\check{z}_{i,2}\hat{\gamma}_iG_2^2 + \frac{1}{2}\check{z}_{i,2}^3$$
$$\dot{\tau}_{i,2} = \check{z}_{i,2}\Psi_{i,2}. \tag{29}$$

Then, the time derivative of $V_{i,2}$ becomes

$$\dot{V}_{i,2} \le -\big(\lambda_{\min}(Q_i) - 4\big)\big\|e_i\big\|^2 + \|P_i\|^2\theta_i + \|P_i\|^2\bar{\delta}_{i,0}^2 + \big\|P_i\big\|^2\big\|L_i\big\|^2\bar{\omega}_1^2 - \sum_{m=1}^{2}k_{i,m}z_{i,m}^2 + \frac{1}{2}z_{i,3}^2 + \sum_{m=1}^{2}\big(g_{i,m}(t)N + 1\big)\dot{\tau}_{i,m} + s_{i,2} + \sum_{m=1}^{2}\frac{\chi_{i,m}^2}{2} + \frac{3}{2} + \tilde{\theta}_i\Big(\pi_{i,2} - \dot{\hat{\theta}}_i\Big) + \tilde{\gamma}_i\Big(\kappa_{i,1} - \dot{\hat{\gamma}}_i\Big) + k_{i,0}\hat{\theta}_i\tilde{\theta}_i - z_{i,2}\lambda\frac{\partial\alpha_{i,1}}{\partial\hat{\theta}_i}\Big(\dot{\hat{\theta}}_i - \pi_{i,2}\Big) + \omega\hat{\gamma}_i\tilde{\gamma}_i + r\hat{\zeta}_i\tilde{\zeta}_i \tag{30}$$

where

$$\pi_{i,2} = \sigma_{i,2} + \pi_{i,1},\ \sigma_{i,2} = \frac{1}{2}\check{z}_{i,2}^2\xi_{i,2}^T\Big(\check{M}_{i,2}\Big)\xi_{i,2}\Big(\check{M}_{i,2}\Big)$$
$$\kappa_{i,1} = \eta_{i,1} - \omega\hat{\gamma}_i,\ \eta_{i,1} = \frac{1}{2}\check{z}_{i,2}^2G_2^2 \tag{31}$$

and $g_{i,2}(t) = \lambda^2$, $s_{i,2} = ([\lambda_M^2\bar{\delta}_{i,1}^2]/2) + (\bar{\delta}_{i,2}^2/2)$, $\chi_{i,2} = (1/2)\theta_i(\xi_{i,2}^T(\bar{M}_{i,2})\xi_{i,2}(\bar{M}_{i,2}) - \xi_{i,2}^T(\check{M}_{i,2})\xi_{i,2}(\check{M}_{i,2}))$ with $\check{M}_{i,2} = [\check{x}_{i,1}, \check{x}_{i,2}, \check{x}_{q,1}]^T$.

Remark 2: So far, the first two steps of backstepping procedure are given, including the design of virtual controllers $\alpha_{i,1}$ and $\alpha_{i,2}$. The next step j $(3 \le j \le n-1)$ is similar to the previous two steps, which mainly including the design of virtual control $\alpha_{i,j}$ and the update of adaptive parameters $\hat{\theta}_i$ and $\hat{\gamma}_i$.

Step j (3 ≤ j ≤ n − 1): Consider the following Lyapunov function candidate:

$$V_{i,j} = V_{i,j-1} + \tfrac{1}{2}z_{i,j}^2. \tag{32}$$

Then

$$\dot{V}_{i,j} = \dot{V}_{i,j-1} + z_{i,j}\Bigg(z_{i,j+1} + \lambda\alpha_{i,j} + \lambda l_{i,j}\big(\check{x}_{i,1} - \hat{x}_{i,1}\big) - \dot{\lambda}\alpha_{i,j-1} - \lambda\Bigg(\sum_{m=1}^{j-1}\frac{\partial\alpha_{i,j-1}}{\partial\tau_{i,m}}\dot{\tau}_{i,m} + \frac{\partial\alpha_{i,j-1}}{\partial\hat{\theta}_i}\dot{\hat{\theta}}_i + \frac{\partial\alpha_{i,j-1}}{\partial\hat{\zeta}_i}\dot{\hat{\zeta}}_i + \frac{\partial\alpha_{i,j-1}}{\partial\hat{\gamma}_i}\dot{\hat{\gamma}}_i + \sum_{m=1}^{j-1}\frac{\partial\alpha_{i,j-1}}{\partial\check{x}_{i,m}}\frac{\lambda\dot{x}_{i,m} - \dot{\lambda}x_{i,m}}{\lambda^2} + \sum_{m=1}^{j-1}\frac{\partial\alpha_{i,j-1}}{\partial\check{x}_{q,m}}\frac{\lambda\dot{x}_{q,m} - \dot{\lambda}x_{q,m}}{\lambda^2}\Bigg) + \dot{\lambda}\hat{x}_{i,j}\Bigg). \tag{33}$$

Similar to the step 2, we have $z_{i,j}\dot{\lambda}(\alpha_{i,j-1} - \sum_{m\equiv1}^{j-1}([\partial\alpha_{i,j-1}/\partial\check{x}_{i,m}]\check{x}_{i,m} - [\partial\alpha_{i,j-1}/\partial\check{x}_{q,m}]\check{x}_{q,m}) + \hat{x}_{i,j}) \le (1/2)\check{z}_{i,j}^2\dot{\lambda}^2\lambda_M^2(\alpha_{i,1} - \sum_{m=1}^{j-1}([\partial\alpha_{i,1}/\partial\check{x}_{i,1}]\check{x}_{i,1} - [\partial\alpha_{i,1}/\partial\check{x}_{j,1}]\check{x}_{j,1}) + \hat{x}_{i,j})^2 + 1/2$, and $z_{i,j}(\sum_{m=1}^{j-1}([\partial\alpha_{i,j-1}/\partial\check{x}_{i,m}]f_{i,m} + [\partial\alpha_{i,j-1}/\partial\check{x}_{q,m}]f_{q,m})) \le (1/2)\check{z}_{i,j}^2\theta_i\xi_{i,j}^T(\bar{M}_{i,j})\xi_{i,j}(\bar{M}_{i,j}) + (1/2) + (z_{i,j}^2/2) + (\bar{\delta}_{i,j}^2/2)$ with $\bar{M}_{i,j} = [\bar{x}_{i,j}, \bar{x}_{q,j-1}]^T$ and $\phi_{i,j} = \lambda_M^2\|\Phi_{i,j}\|^2$, $\delta_{i,j}$ is the approximation error, $|\delta_{i,j}| \le \bar{\delta}_{i,j}$.

The virtual controller $\alpha_{i,j}$ is chosen as

$$\alpha_{i,j} = -k_{i,j}\check{z}_{i,j} - \frac{3}{2}\check{z}_{i,j} + \sum_{m=1}^{j-1}\frac{\partial\alpha_{i,j-1}}{\partial\tau_{i,m}}\dot{\tau}_{i,m} + \frac{\partial\alpha_{i,j-1}}{\partial\hat{\theta}_i}\pi_{i,j} + \frac{\partial\alpha_{i,j-1}}{\partial\hat{\zeta}_i}\dot{\hat{\zeta}}_i + \sum_{m=2}^{j-1}\check{z}_{i,m}\frac{\partial\alpha_{i,m-1}}{\partial\hat{\theta}_i}\sigma_{i,j} + \frac{\partial\alpha_{i,j-1}}{\partial\hat{\gamma}_i}\kappa_{i,j-1} + \sum_{m=3}^{j-1}\check{z}_{i,m}\frac{\partial\alpha_{i,m-1}}{\partial\hat{\gamma}_i}\eta_{i,j-1} + \sum_{m=1}^{j-1}\frac{\partial\alpha_{i,j-1}}{\partial\check{x}_{i,m}}\check{x}_{i,m+1} + \sum_{m=1}^{j-1}\frac{\partial\alpha_{i,j-1}}{\partial\check{x}_{q,m}}\check{x}_{q,m+1} - l_{i,j}\big(\check{x}_{i,1} - \hat{x}_{i,1}\big) + N\big(\tau_{i,j}\big)\Psi_{i,j} \tag{34}$$

with

$$\Psi_{i,j} = \frac{1}{2}\check{z}_{i,j}\hat{\theta}_i\xi_{i,j}^T\Big(\check{M}_{i,j}\Big)\xi_{i,j}\Big(\check{M}_{i,j}\Big) + \frac{1}{2}\check{z}_{i,j}\hat{\gamma}_iG_j^2 + \frac{1}{2}\check{z}_{i,j}^3$$
$$\dot{\tau}_{i,j} = \check{z}_{i,j}\Psi_{i,j}. \tag{35}$$

Then, the time derivative of $V_{i,j}$ becomes

$$\dot{V}_{i,j} \le -\big(\lambda_{\min}(Q_i) - 4\big)\big\|e_i\big\|^2 + \|P_i\|^2\theta_i + \|P_i\|^2\bar{\delta}_{i,0}^2 + \big\|P_i\big\|^2\big\|L_i\big\|^2\bar{\omega}_1^2 - \sum_{m=1}^{j}k_{i,m}z_{i,m}^2 + \frac{1}{2}z_{i,j+1}^2 + \sum_{m=1}^{j}\big(g_{i,m}(t)N + 1\big)\dot{\tau}_{i,m} + \sum_{m=1}^{j}\frac{\chi_{i,m}^2}{2} + s_{i,j}$$

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TABLE III
DESIGN OF CONTROLLERS AND ADAPTIVE LAWS FOR STEP n

Control Laws:

$$
\begin{aligned}
u_i &= -k_{i,n}\check{z}_{i,n} - \check{z}_{i,n} + \textstyle\sum_{m=1}^{n-1} \frac{\partial \alpha_{i,n-1}}{\partial \tau_{i,m}} \dot{\tau}_{i,m} + \frac{\partial \alpha_{i,n-1}}{\partial \hat{\theta}_i} \pi_{i,n} \\
&\quad + \textstyle\sum_{m=2}^{n-1} \check{z}_{i,m} \frac{\partial \alpha_{i,m-1}}{\partial \hat{\theta}_i} \sigma_{i,n} + \frac{\partial \alpha_{i,n-1}}{\partial \hat{\gamma}_i} \pi_{i,n-1} + \frac{\partial \alpha_{i,n-1}}{\partial \hat{\zeta}_i} \dot{\hat{\zeta}}_i \\
&\quad + \textstyle\sum_{m=3}^{n-1} \check{z}_{i,m} \frac{\partial \alpha_{i,m-1}}{\partial \hat{\gamma}_i} \eta_{i,n-1} + \sum_{m=1}^{n-1} \frac{\partial \alpha_{i,n-1}}{\partial \bar{x}_{i,m}} \check{x}_{i,m+1} \\
&\quad + \textstyle\sum_{m=1}^{n-1} \frac{\partial \alpha_{i,n-1}}{\partial \bar{x}_{q,m}} \check{x}_{q,m+1} - l_{i,n}(\check{x}_{i,1} - \hat{x}_{i,1}) \\
&\quad + N(\tau_{i,n})\Psi_{i,n}
\end{aligned}
$$

with

$$
\begin{aligned}
&\Psi_{i,n} = \tfrac{1}{2}\check{z}_{i,n}\hat{\theta}_i \xi_{i,n}^T(\check{M}_{i,n})\xi_{i,n}(\check{M}_{i,n}) + \tfrac{1}{2}\check{z}_{i,n}\hat{\gamma}_i G_n^2 + \tfrac{1}{2}\check{z}_{i,n}^3 \\
&\dot{\tau}_{i,n} = \check{z}_{i,n}\Psi_{i,n} \\
&\pi_{i,n} = \check{z}_{i,n}\sigma_{i,n} + \pi_{i,n-1} \\
&\sigma_{i,n} = \tfrac{1}{2}\check{z}_{i,n}\xi_{i,n}^T(\check{M}_{i,n})\xi_{i,n}(\check{M}_{i,n}) \\
&\kappa_{i,n-1} = \check{z}_{i,n}\eta_{i,n-1} + \kappa_{i,n-2} \\
&\eta_{i,n-1} = \tfrac{1}{2}\check{z}_{i,n}^2 G_n^2
\end{aligned}
\tag{S1}
$$

Adaptive Laws:

$$
\begin{aligned}
&\dot{\hat{\gamma}}_i = \kappa_{i,n-1} \\
&\dot{\hat{\theta}}_i = \pi_{i,n}
\end{aligned}
\tag{S2}
$$

$$
\begin{aligned}
&+ \frac{2j-1}{2} + \tilde{\theta}_i\Big(\pi_{i,j} - \dot{\hat{\theta}}_i\Big) + \tilde{\gamma}_i\Big(\kappa_{i,j-1} - \dot{\hat{\gamma}}_i\Big) + k_{i,0}\hat{\theta}_i\tilde{\theta}_i \\
&- \lambda \sum_{m=2}^{j} z_{i,m} \frac{\partial \alpha_{i,m-1}}{\partial \hat{\theta}_i}\Big(\dot{\hat{\theta}}_i - \pi_{i,j}\Big) + \omega\hat{\gamma}_i\tilde{\gamma}_i + r\hat{\zeta}_i\tilde{\zeta}_i \\
&- \lambda \sum_{m=3}^{j} z_{i,m} \frac{\partial \alpha_{i,m-1}}{\partial \hat{\gamma}_i}\Big(\dot{\hat{\gamma}}_i - \kappa_{i,j-1}\Big)
\end{aligned}
\tag{36}
$$

where $\pi_{i,j} = \check{z}_{i,j}\sigma_{i,j} + \pi_{i,j-1}$, $\sigma_{i,j} = (1/2)\check{z}_{i,j}\xi_{i,j}^T(\check{M}_{i,j})\xi_{i,j}(\check{M}_{i,j})$, $\kappa_{i,j-1} = \check{z}_{i,j}\eta_{i,j-1} + \kappa_{i,j-2}$, $\eta_{i,j-1} = (1/2)\check{z}_{i,j}^2 G_j^2$, $G_j = \alpha_{i,j-1} - \sum_{m=1}^{j-1}([\partial\alpha_{i,j-1}]/[\partial\check{x}_{i,m}]\check{x}_{i,m} + [\partial\alpha_{i,j-1}]/[\partial\check{x}_{q,m}]\check{x}_{q,m}) + \hat{x}_{i,j}$, $g_{i,j}(t) = \lambda^2$, $s_{i,j} = ([\lambda_M^2\bar{\delta}_{i,1}^2]/2) + \sum_{m=2}^{j}([\bar{\delta}_{i,m}^2]/2)$, $\chi_{i,j} = (1/2)\theta_i(\xi_{i,j}^T(\bar{M}_{i,j})\xi_{i,j}(\bar{M}_{i,j}) - \xi_{i,j}^T(\check{M}_{i,j})\xi_{i,j}(\check{M}_{i,j}))$ with $\check{M}_{i,j} = [\bar{\check{x}}_{i,j}, \bar{\check{x}}_{q,j-1}]^T$.

Step n: Consider the following Lyapunov function candidate:

$$
V_{i,n} = V_{i,n-1} + \frac{1}{2}z_{i,n}^2. \tag{37}
$$

By following the ideas in the above step q ($q = 1, 2, \ldots,$ n-1), the final controllers and adaptive laws are obtained recursively, as summarized in Table III, where $G_n = -\alpha_{i,n-1} + \sum_{m=1}^{n-1}([\partial\alpha_{i,n-1}]/[\partial\check{x}_{i,m}]\check{x}_{i,m} + [\partial\alpha_{i,n-1}]/[\partial\check{x}_{q,m}]\check{x}_{q,m}) + \hat{x}_{i,n}$.

Then, the time derivative of $V_{i,n}$ becomes

$$
\begin{aligned}
\dot{V}_{i,n} \le& -\big(\lambda_{\min}(Q_i) - 4\big)\|e_i\|^2 + \|P_i\|^2\theta_i + \|P_i\|^2\bar{\delta}_{i,0}^2 \\
&+ \|P_i\|^2\|L_i\|^2\bar{\omega}_1^2 - \sum_{m=1}^{n} k_{i,m}z_{i,m}^2 \\
&+ \sum_{m=1}^{n}\big(g_{i,m}(t)N + 1\big)\dot{\tau}_{i,m} + \sum_{m=1}^{n}\frac{\chi_{i,m}^2}{2} + s_{i,n} \\
&+ \frac{2n-1}{2} + k_{i,0}\hat{\theta}_i\tilde{\theta}_i + \omega\hat{\gamma}_i\tilde{\gamma}_i + r\hat{\zeta}_i\tilde{\zeta}_i
\end{aligned}
\tag{38}
$$

where $g_{i,n}(t) = \lambda^2$, $s_{i,n} = ([\lambda_M^2\bar{\delta}_{i,1}^2]/2) + \sum_{m=2}^{n}([\bar{\delta}_{i,m}^2]/2)$, $\chi_{i,n} = (1/2)\theta_i(\xi_{i,n}^T(\bar{M}_{i,n})\xi_{i,n}(\bar{M}_{i,n}) - \xi_{i,n}^T(\check{M}_{i,n})\xi_{i,n}(\check{M}_{i,n}))$ with $\check{M}_{i,n} = [\bar{\check{x}}_{i,n}, \bar{\check{x}}_{q,n-1}]^T$.

C. System Stability

In order to analyze the stability of the proposed distributed control strategy, the total Lyapunov function V is constructed as

$$
V = V_{i,n}. \tag{39}
$$

Now, the following theorems can be concluded as follows.

Theorem 1: For a nonlinear MASs (1) suffering from unknown sensor attacks. Suppose that Assumptions 1–3 hold, if the controller (S1), the virtual controller (22), (28), and (34) are applied, then all the signals of the closed-loop systems are bounded and the bipartite tracking error can converge to an adjustable small neighborhood of origin.

Proof: By using Assumption 3 and Lemma 3, $\chi_{i,m}$ is bounded as $|\chi_{i,m}| \le \bar{\chi}_{i,m}$, where $\bar{\chi}_{i,m} > 0$ is a constant. From $k_{i,0}\hat{\theta}_i\tilde{\theta}_i \le -(k_{i,0}/2)\tilde{\theta}_i^2 + (k_{i,0}/2)\theta_i^2$, $\omega\hat{\gamma}_i\tilde{\gamma}_i \le -(\omega/2)\tilde{\gamma}_i^2 + (\omega/2)\gamma_i^2$, $r\hat{\zeta}_i\tilde{\zeta}_i \le -(r/2)\tilde{\zeta}_i^2 + (r/2)\zeta_i^2$. Then, we have

$$
\begin{aligned}
\dot{V} \le& -\big(\lambda_{\min}(Q_i) - 4\big)\|e_i\|^2 + \|P_i\|^2\theta_i + \|P_i\|^2\bar{\delta}_{i,0}^2 \\
&+ \|P_i\|^2\|L_i\|^2\bar{\omega}_1^2 - \sum_{m=1}^{n} k_{i,m}z_{i,m}^2 + \sum_{m=1}^{n}\frac{\bar{\chi}_{i,m}^2}{2} + s_{i,n} \\
&+ \sum_{m=1}^{n}\big(g_{i,m}(t)N(\tau_{i,m}) + 1\big)\dot{\tau}_{i,m} - \frac{k_{i,0}}{2}\tilde{\theta}_i^2 - \frac{\omega}{2}\tilde{\gamma}_i^2 \\
&- \frac{r}{2}\tilde{\zeta}_i^2 + \frac{2n-1}{2} + \frac{k_{i,0}}{2}\theta_i^2 + \frac{\omega}{2}\gamma_i^2 + \frac{r}{2}\zeta_i^2.
\end{aligned}
\tag{40}
$$

Denote $p_i = \min\{([\lambda_{min}(Q_i) - 4]/[\lambda_{\max}(P_i)]), 2k_{i,1}\lambda_M/\lambda^2, 2\ k_{i,2}\ , \ldots, 2k_{i,n}, k_{i,0}, \omega, r\}$, $\nu_i = \sum_{m=1}^{n}(\bar{\chi}_{i,m}^2/2) + s_{i,n} + (2n - 1/2) + (k_{i,0}/2)\theta_i^2 + (\omega/2)\gamma_i^2 + (r/2)\zeta_i^2 + \|P_i\|^2\theta_i + \|P_i\|^2\bar{\delta}_{i,0}^2 + \|P_i\|^2\|L_i\|^2\bar{\omega}_1^2$. Then, we have

$$
\dot{V} \le -p_i V + \nu_i + \sum_{m=1}^{n}\big(g_{i,m}(t)N(\tau_{i,m}) + 1\big)\dot{\tau}_{i,m}. \tag{41}
$$

Furthermore, the following inequality holds:

$$
V(t) \le C_0 + e^{-p_i t}\int_0^t \sum_{m=1}^{n}\big(g_{i,m}(t)N(\tau_{i,m}) + 1\big)\dot{\tau}_{i,m}e^{p_i\tau}d\tau \tag{42}
$$

where $C_0 = V(0) + (\nu_i/p_i)$. By analyzing the above inequality and choose $\lambda_{min}(Q_i) - 4 > 0$, all closed-loop signals are bounded. When $t \to \infty$, the local tracking error $z_{i,1}$ will converge to

$$
|z_{i,1}| \le H_i = \sqrt{\frac{2\nu_i}{p_i}}. \tag{43}
$$

From $z_{i,1} = x_{i,1} - \text{sign}(a_{ij})x_{j,1}$ and the definition of a shortest path S_i between the node o_i and the leader y_d ($S_i : y_d \to o_{j_k} \ldots \to o_{j_1} \to o_j \to o_i$), we have

$$
\begin{aligned}
z_{i,1} &= x_{i,1} - \text{sign}\big(a_{i,j}\big)x_{j,1} \\
&= x_{i,1} - \text{sign}\big(a_{i,j}\big)\big(z_{j,1} + \text{sign}\big(a_{j,j_1}\big)\big(z_{j_1,1} + \text{sign}\big(a_{j_1,j_2}\big) \\
&\qquad \big(\ldots\big(z_{j_k,1} + \text{sign}\big(a_{j_k,d}\big)y_d\big)\big)\big)\big)
\end{aligned}
$$

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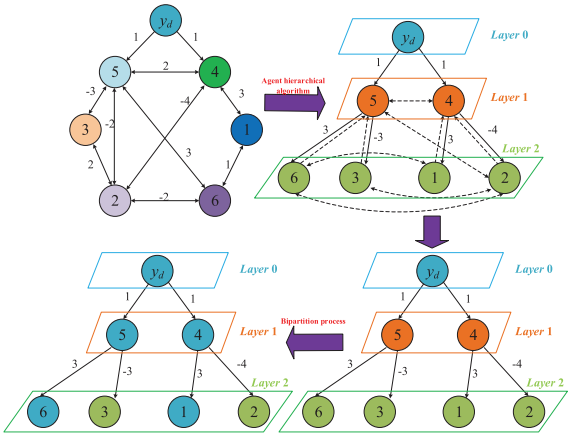


Fig. 3. Evolution of connected structurally balanced topology.

$$= x_{i,1} - \text{sign}(a_{i,j})z_{j,1} - \text{sign}(a_{i,j})\text{sign}(a_{j,j_1})z_{j_1,1} - \ldots - \text{sign}(a_{i,j})\text{sign}(a_{j,j_1})\ldots\text{sign}(a_{j_{k-1},j_k})z_{j_k,1} - \text{sign}(S_i)y_d. \tag{44}$$

From (44), it can be seen that

$$\begin{aligned} |\Lambda_i| &= |x_{i,1} - \text{sign}(S_i)y_d| \\ &\le |z_{i,1}| + |z_{j,1}| + |z_{j_1,1}| + \ldots + |z_{j_k,1}| \\ &\le \sum_{c\in\{k|o_k\in S_i\}} H_c \end{aligned} \tag{45}$$

where $\sum_{c\in\{k|o_k\in S_i\}} H_c = \sqrt{(2\nu_i/p_i)} + \sqrt{(2\nu_j/p_j)} + \sqrt{(2\nu_{j_1}/p_{j_1})} + \cdots + \sqrt{(2\nu_{j_k}/p_{j_k})}$. From (45), the bipartite tracking error Λ_i will converge to $\sum_{c\in\{k|o_k\in S_i\}} H_c$ as $t \to \infty$. This completes the proof of Theorem 1. ■

Remark 3: Based on the analysis of the compact set H_i in (43), the error $z_{i,1}$ is affected by $k_{i,m}$, $k_{i,0}$, ω, $\bar{\chi}_{i,m}$, $\bar{\delta}_{i,m}$, θ_i, and γ_i. Thus, the error can be adjusted by reducing $k_{i,0}$, ω, and increasing $k_{i,m}$. Therefore, by selecting appropriate design parameters, the tracking error can converge to an adjustable small neighborhood of origin.

Remark 4: It should be mentioned that the bipartition process can also be achieved for the structurally balanced graph. For example, Fig. 3 is a structurally balanced graph and the shortest paths for nodes $o_1 - o_6$ are $S_1 : y_d \to o_4 \to o_1$, $S_2 : y_d \to o_4 \to o_2$, $S_3 : y_d \to o_5 \to o_3$, $S_4 : y_d \to o_4$, $S_5 : y_d \to o_5$, and $S_6 : y_d \to o_5 \to o_6$, respectively. Besides, $S_1 > 0$, $S_2 < 0$, $S_3 < 0$, $S_4 > 0$, $S_5 > 0$, $S_6 > 0$. Then, two different sets can be obtained, that is, $M_1 = \{y_d, o_1, o_4, o_5, o_6\}$ and $M_2 = \{o_2, o_3\}$.

Remark 5: In the existing research results [44], the term $1/\check{z}_i \sum_{p=1}^{i-2} \sum_{m=p+1}^{i-1} \check{z}_m[\partial\alpha_{m-1}/\partial\hat{\lambda}_p]\omega_{p,i-p+1}$ appeared in the process of virtual and actual control signals in step i, $3 \le i \le n$. However, singularity may arise from the term $1/\check{z}_i$. To handle this challenge, we construct a new auxiliary variable as $\pi_{i,j} = \check{z}_{i,j}\sigma_{i,j} + \pi_{i,j-1}$. In this way, our proposed virtual controller and distributed security control scheme can avoid the singularity problem in the process of adaptive iteration in this article. In addition, in the traditional backstepping design, the derivative of the virtual control of step n will need to rely on the information of the neighbor's control input u_j ($[\partial\alpha_{n-1}/\partial x_{j,n}]u_j$ with u_j being neighbor's control signal), which will lead to the algebraic loop problem. In this article, the hierarchical algorithm is presented to ensure that each agent only receives the information from the upper layer such that this problem can be overcame.

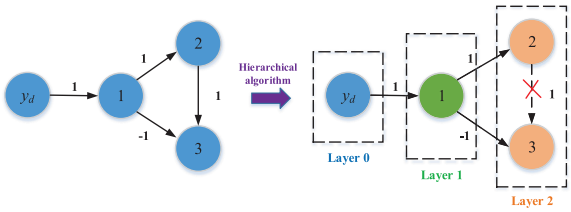


Fig. 4. Communications structurally unbalanced topology.

IV. Simulation Studies

The bipartite consensus tracking problem for a set of forced damped pendulums (FDPs) with sensor attacks is simulated in this section, which can be described by [50]

$$\ddot{\vartheta}_i = \frac{b_i}{m_i l_i^2}\dot{\vartheta}_i - \frac{c_i g}{l_i}\sin(\vartheta_i) + u_i, \; i = 1, 2, 3 \tag{46}$$

where $g = 9.8$, $l_i = 1$, $b_i = -0.25$, and $c_i = 9.8^{-1}$. Let $x_{i,1} = \vartheta_i$, $x_{i,2} = \dot{\vartheta}_i$, then the system model can be rewritten as

$$\begin{aligned} \dot{x}_{i,1} &= x_{i,2} \\ \dot{x}_{i,2} &= -\frac{b_i}{m_i l_i^2}x_{i,2} - \frac{c_i g}{l_i}\sin(x_{i,1}) + u_i \\ y_i &= x_{i,1}, \; i = 1, 2, 3. \end{aligned} \tag{47}$$

The leader's dynamic equation is

$$\begin{aligned} \dot{x}_d &= -1.2x_d + 0.5\cos\left(\frac{t}{2} + 3\right) + 3t^2e^{-t} - 0.5 \\ y_d &= x_d. \end{aligned} \tag{48}$$

The communication graphs are considered in Fig. 4. As shown in Fig. 4, since the communication graphs is structurally unbalanced, the hierarchical algorithm is first applied to divide all agents into two different groups. From Fig. 4, the shortest routes for agents $o_1 - o_3$ are $S_1 : y_d \to o_1$, $S_2 : y_d \to o_1 \to o_2$, and $S_3 : y_d \to o_1 \to o_3$, respectively. Besides, $S_1 > 0$, $S_2 > 0$, $S_3 < 0$. Then, $M_1 = \{y_d, o_1, o_2\}$ and $M_2 = \{o_3\}$ can be given.

In simulation, the initial states of the three follower agents are selected as $x_1(0) = [0.1, 0]^T$, $x_2(0) = [0.2, 0.1]^T$, $x_3(0) = [0.3, 0.2]^T$ for all agents. The attacks are chosen as $\rho(t) = 1 + 0.25\cos(t)$. The design parameters are chosen as $k_{i,1} = 10$, $k_{i,2} = 1$. $k_{i,0} = 0.01$, $\omega = 0.01$, $N(\tau_{i,j}) = e^{\tau_{i,j}^2}\sin(\pi\tau_{i,j})$ for $i = 1, \ldots, 3$ and $j = 1, 2$.

Simulation results are displayed in Figs. 5–9, respectively. Fig. 5 shows output trajectories of all agents. The distributed bipartite consensus errors are pictured in Fig. 6. The adaptive parameters are pictured in Figs. 7–8. Fig. 9 displays the control input of $u_i (i = 1, \ldots, 3)$ under the proposed protocol.

V. Conclusion

This study has presented a novel adaptive bipartite consensus tracking strategy for MASs under sensor deception

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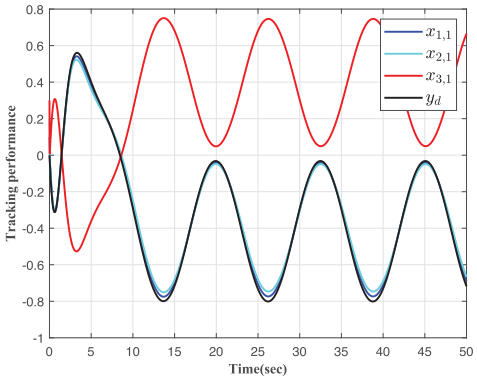


Fig. 5. Bipartite consensus performance.

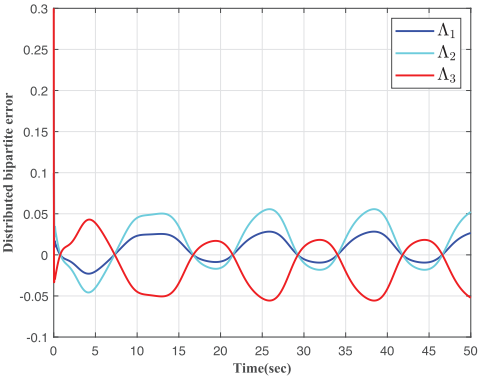


Fig. 6. Distributed bipartite errors.

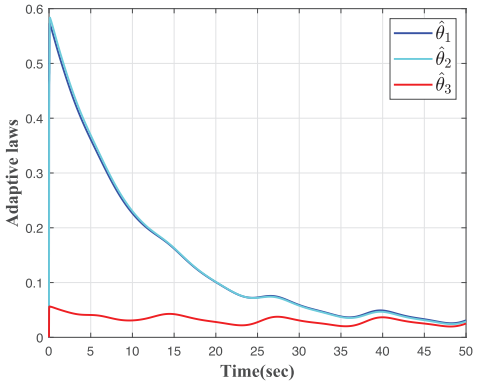


Fig. 7. Trajectory of the adaptive parameters $\hat{\theta}_i (i = 1, 2, 3)$.

attacks. The innovation of this article is that the consensus tracking problem for multiagents is decomposed into the tracking problem for a single agent by using a hierarchical algorithm. Based on this technique, the distributed security control algorithm is able to present by constructing a new coordinate transformation involving the attacked and estimated state variables, and the singularity problem is avoided gracefully. The proposed method ensures that the distributed consensus tracking errors converge to a small neighborhood of the origin, and all the signals of the closed-loop system remain bounded. In the future, to save communication and computational resources, our attention will be focused on a more challenging issue where the compromised output signal is transmitted through an event-triggered mechanism, that is,

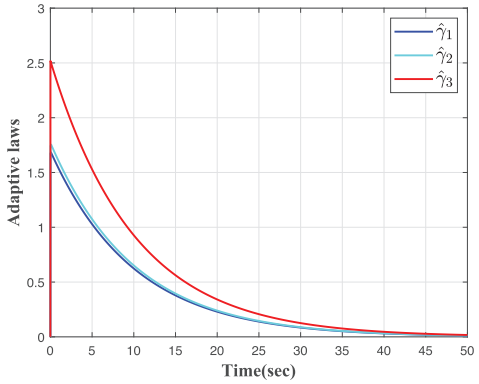


Fig. 8. Trajectory of the adaptive parameters $\hat{\gamma}_i (i = 1, 2, 3)$.

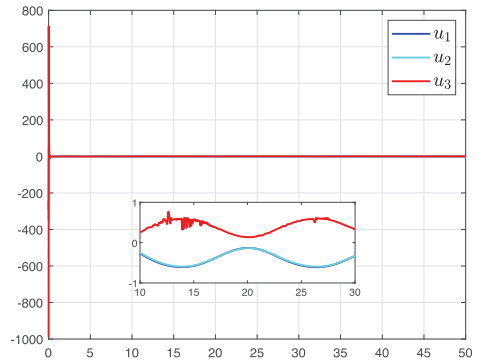


Fig. 9. Response of the control input $u_i (i = 1, 2, 3)$.

the intermittent $\bar{\bar{x}}_{i,1}$, rather than $\check{x}_{i,1}$ is available for the feedback design. In such a case, how to develop the effective state observer and distributed security control strategy for nonlinear MASs are not a trivial matter.

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